

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
9.	(a)			Indicative content 1. Oxidation states in Group 4 are stable at +4 at the top of the group and +2 at the bottom 2. Due to the inert pair effect 3. The inert pair effect increases down the group 4. PbO more stable than PbO₂ / PbO₂ is an oxidising agent / CO₂ more stable than CO / CO is a reducing agent 5. SiO doesn't form as this is a +2 oxidation state which cannot form this high in the group 6. +5 oxidation state possible in third period of Group 5 but not in Period 2 7. This is because phosphorus can expand its octet / phosphorus has available d-orbitals 8. NCl₅ not possible as N has no available d-orbitals so it cannot expand its octet 9. One characteristic of transition metals is that they form a variety of oxidation states 10. Mn can form a range of oxidation states as the 3d orbitals (and 4s) have similar energies and similar ionisation energies 5-6 marks: At least seven relevant points, discussing the oxidation states of all three sets of elements <i>The candidate constructs a relevant, coherent and logically structured account including all key elements of the indicative content, clearly showing an understanding of the reasons for the differing oxidation states in all three categories. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i>	6	4	2			